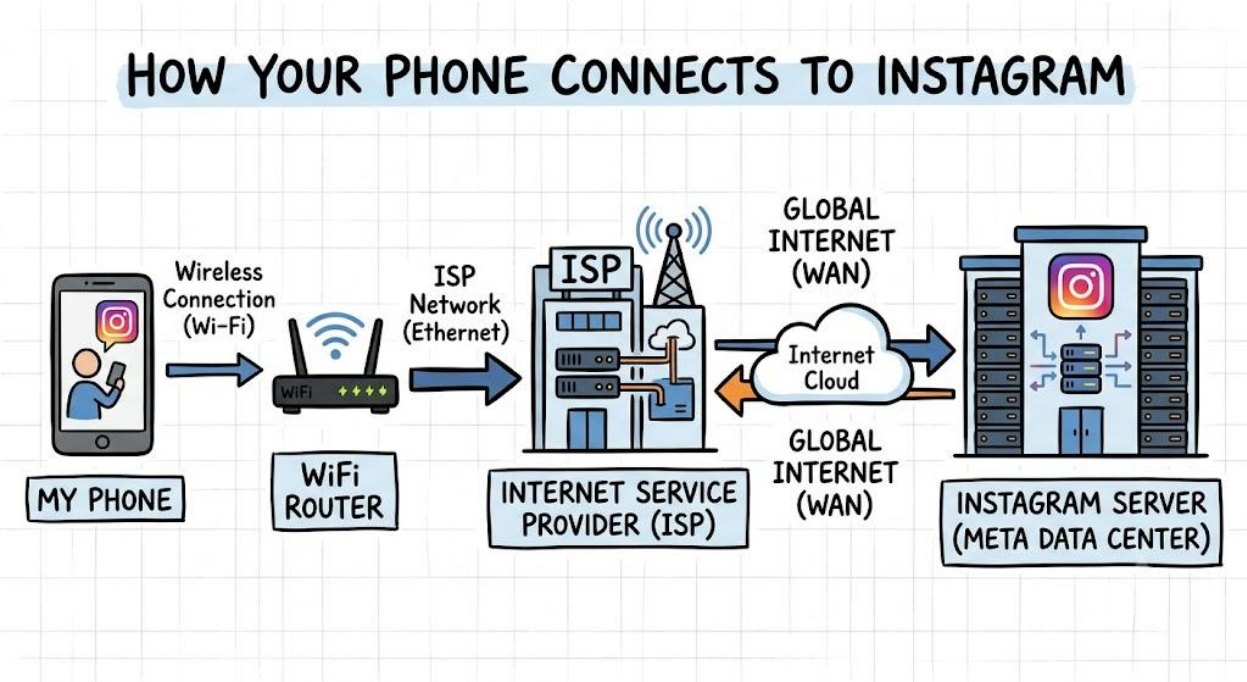
INTRODUCTION TO NETWORK

1. Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.

Ans:-



- **My Phone to Wi-Fi Router**: A wireless connection (Wi-Fi) links your device to your home or local router.
- **Wi-Fi Router to ISP**: A wired ISP network (Ethernet) transfers the data from your router to your Internet Service Provider.
- **ISP to Global Internet (WAN)**: The ISP sends the data out through the Internet cloud and global WAN infrastructure.
- **Instagram Server (Meta Data Center)**: Finally, the request reaches the Instagram servers, allowing the application to load your feed, messages, and media.

2. List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smart watch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

Ans:-

No.	Step / Component	Network / Connection Type	Description
1	**Smartphone to Wi-Fi Router**	WLAN (Wireless Local Area Network)	Connects wirelessly to the home or local Wi-Fi router via Wi-Fi signals.
2	**Wi-Fi Router to ISP**	LAN / Ethernet	Routes local network data through a physical Ethernet cable/broadband line to the Internet Service Provider.
3	**ISP to Global Internet**	WAN (Wide Area Network)	Transfers data across the internet service provider's network infrastructure and global internet routes.
4	**Global Internet to Instagram Server**	WAN / Data Center Network	Delivers requests to Meta's server infrastructure to process requests and fetch feed content.

3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

Ans:-

- **Internet:** The Internet is a public network that anyone can access with an internet connection. For example, I use the Internet to watch YouTube videos and browse websites.

- **Intranet:** An intranet is a private network used within an organization. For example, my college may use an intranet to share notices, attendance records, and study materials with students and staff.

Difference: The Internet is publicly accessible, while an intranet is restricted to authorized users.

Answer:

- **Internet:** A global public network that anyone can access. Example: public WiFi at a café or railway station, where we can open Google, YouTube or Instagram.

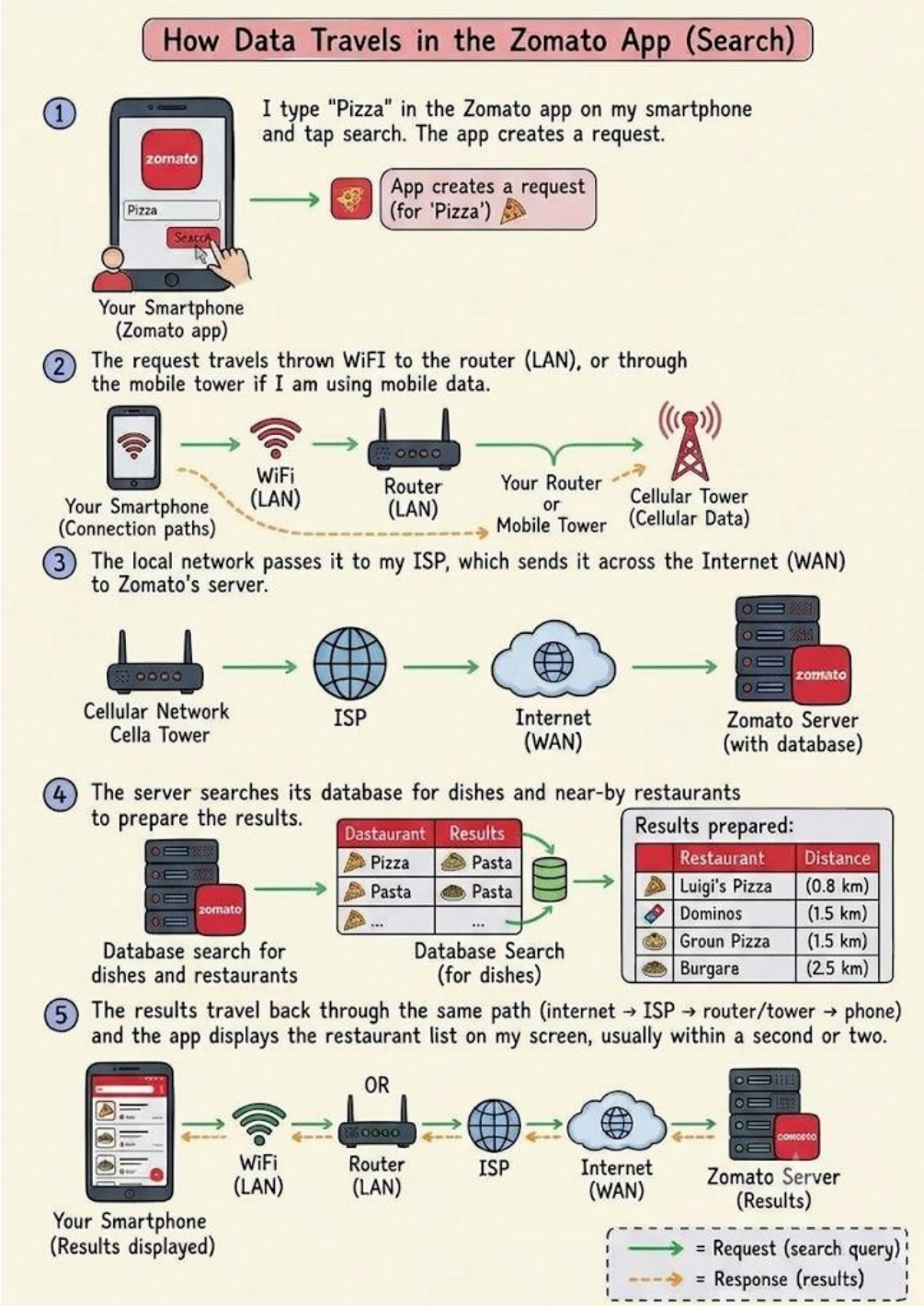
- **Intranet:** A private network used only inside an organization by authorized people. Example: the college portal, accessible only through college WiFi or student login. It is used for attendance, results, notes and the library.

Point	Internet	Intranet
Access	Public, open to everyone	Private, only for members
Size	Worldwide	Limited to one organization
Security	Less secure	More secure (login, firewall)
Example	Google, public WiFi	College portal, office network

4. Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

Hint: Mention the devices and networks involved, and what happens when you hit 'search'.

Ans:-



- **Step 1 (Request Created):** You type a search query (e.g., "Pizza") in the app on your smartphone, which generates a digital data request.

- **Step 2 (Local Network):** The request travels wirelessly from your phone to your local Wi-Fi router (LAN) or a nearby mobile cell tower.

- **Step 3 (ISP & Internet):** Your local network passes the request to your Internet Service Provider (ISP), which routes it across the global Internet (WAN) to the app's remote server.

- **Step 4 (Server Processing):** The server processes your request, searches its database for matching information (like nearby restaurants or items), and prepares the results.

- **Step 5 (Response Delivered):** The response travels back along the reverse path ($\text{Internet} \rightarrow \text{ISP} \rightarrow \text{Router/Tower} \rightarrow \text{Smartphone}$), displaying the results on your screen within seconds.